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Work motivation and level of performance:
A disappointing relationship?

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Presented in Symposium:
Integrative Approaches to Work Motivation: Ability and Non-ability Determinants of Regulatory Processes, Learning, and Performance
Organized by Ruth Kanfer (Georgia Institute of Technology, Atlanta, USA)
XXV International Congress of Applied psychology
Singapore, July 6th -12th, 2002

Abstract

From a common sense perspective, motivation should be a strong predictor of performance. Yet a perusal of various recent meta-analytic studies on the determinants of performance indicates that motivation rarely is strongly related to performance. Indeed, aptitude consistently outperforms most other predictors. Looking back at intelligence assessment history, teachers in Paris, complained about not having good measures of aptitude to distinguish students that were not motivated enough from those not having the necessary amount of aptitude. This starting point helped Binet to solve the problem with assessing intelligence. Why have psychologists not been successful in developing equally powerful assessments of motivation during the past 100 years?

I suggest that a closer look at the conceptual problems and challenges faced by personality and attitude researchers in their fields may be particularly useful in addressing the motivation-performance issue in organizational psychology. First there is the criterion problem. That is, for fair evaluation of the influence of motivational dispositions, we need to ask whether the performance criteria used are aggregated to the same level of generality. In our laboratory, we refer to this as Brunswik symmetry. Dispositions are tendencies to act, and tendencies are never adequately validated via single behavioral acts, but rather require multiple act criteria. Epstein has emphasized the implications of aggregation against Mischel's verdict concerning personality dispositions. Second, Ajzen and Fishbein note similar problems in attitude research, and formulated the necessity of principles of correspondence in action, target, time and context. Research is reviewed to investigate whether these conceptual strategies help in the use of work motivation measures to predict performance.

Introduction

Good afternoon!

Guess that only a few of you are familiar with my research. No wonder, I have not published a single generic paper on motivation, nor on work motivation. So why am I here? Just consider me as the new kid on the block. New kids are outsiders at the beginning, but they might bring with them some new perspectives, which, I do hope, help to shed some new light on the topics, the old kids on the block deal with.

Actually, the real cause why I am here is Ruth Kanfer. When I was at an international conference on learning and individual differences, organized by Phil Ackerman and

colleagues a few years ago, at Minnesota, I presented the following figure summarizing our research.

Fig.1 about here-

You see that we had been able to predict work performance, defined as a surrogate for real life performance, actually an aggregate consisting out of three different computer-based complex business games, to an impressive degree. The variance accounted for is around 50%. Only the Kanfer/Ackerman lab has similar results and Phil and Ruth are the only ones, who beat us from time to time as regards predictability of performance, e.g. in their research using the air traffic controller scenario (TRACON).

Commenting on our results, Ruth very politely said, Werner nice results, but where is motivation? Well Ruth directly pointed to an Achill's heel of our research. We did not bet and invest on the "will-do" factors. Our main concerns are the "can-do" factors. Actually our lab also has some nice results from an earlier study a couple of our lab members did, while they had been in A.O. Jäger's group in Berlin, with whom we cooperated over many years, when he was still active. Fig.2 shows these results:

Fig.2 about here

Achievement motivation assessed via the Freiburg Personality Inventory (FPI_R) predicted some of our constructs, but had indirect effects on performance only. Please note that a more circumscribed domain specific assessment of motivation, namely the one to deal with computers had a small but interesting incremental direct effect on performance. This complies pretty well with what many of you claim, namely motivation might be more domain specific than the "can-do" constructs.

On the other hand, the whole framework and results of our research fits well into what Phil Ackerman (1996) has introduced as the PPIK- theory. Intelligence as process (working memory and reasoning as fluid intelligence), personality, interest, and motivation and intelligence as knowledge are described in a path-analytic stance. Our results quite clearly bolster such a framework. Intelligence as knowledge has the strongest and best direct effects on performance. Intelligence as process also contributes with direct, but somewhat smaller effects. Yet, its main contribution is indirect via knowledge. Motivation is interesting, but also mainly in its indirect effects on other more important constructs, which mediate their effect? Shall we accept such a conclusion?

Why is there no convincing meta-analysis, demonstrating the predictive impact of motivational constructs?

Going back more than a century to Paris, we heard teachers complaining about the performance of their students. They wanted to have assessments of the "can-do" factors and the "will-do" factors to be able to explain the performances of their students in terms of grades. This brought Alfred Binet onto the stage. He had the tools to assess the "can-do" factors in his hands, which only had to be expanded and tailored to the school system. The hypothesis was, once we know the individual differences in the "can-do" factors, students with high scores on them but bad grades must have deficits in "will-do" factors like motivation and/or interests. Binet's approach was successful and imported by Terman to the United States, and extended by many others over the last century. Given such a large time- period should we not expect similar successes with the assessment of motivation?

Fig. 3 visualizes the success story of intellectual abilities, in predicting work performance not only in school, but in the adult work force as well, with a ballantine. Many meta-analyses of the Hunter and Schmidt (1990) type underscore this fact. Intellectual abilities are the best predictors of performance. Variance accounted for is impressive, but far from being perfect, leaving room for many other constructs. So what about motivation? Fig. 4 shows questions marks and sad state of affairs. Despite single papers hinting to and claiming the importance of need of achievement, need for power and other variants of motivational constructs, like in David McClelland's research, one does not find similar impressive results synthesized in a meta-analysis. Is it, that motivation overlaps with intelligence (Fig. 5) and has it only indirect effects? On the other hand, wait a minute, how far does motivation carry you, if you do not have the right amount of KSA's? Does motivation's importance show up only by truly considering their interaction with ability (see Fig. 6)?

Fig 3 – 6 about here

Yet, research on personality x ability interactions and the related fields of aptitude x treatment interaction (ATI) show disappointing, non-generalizable results as well (Cronbach & Snow, 1977). All these results are at variance with what we expect from a common sense knowledge base. We have many single case examples, which hint to quite the opposite, what would Boris Becker the German tennis champion, Oliver Kahn the German soccer goalie, Tiger Woods the American golf master and countless others have been without motivation? On the other hand, would you be here in Singapore without hard work and strong motivation?

So, what is wrong with our research in these areas?Let me concentrate on four topics and possible reasons:

a) Are we using the wrong theoretical frameworks?
b) Are we looking at the wrong criteria?
c) Are we using the wrong measures and assessment strategies?
d) Psychometric problems in assessing interactions.

The motivation to deal with these topics from a personal stance and via hindsight is fueled I have to confess by a "disease" I suffer now for many years, which might be labeled "science envy". Why are these girls and guys in sciences like physics, chemistry and biology so successful? Some say these fields do not need the same amount of aptitudes as our field, which is much more demanding and complicated. Even scientists from these areas comply occasionally with such a statement.

Ruminations and speculations about the right theoretical framework

Looking for reasons and a possible treatment against "science envy" we find the application of principles of symmetry as key elements in explaining their success (Zee, 1989). Can psychology not also capitalize on such principles for its theoretical frameworks? Yes, it can, and these principles already have been delivered in the contributions of Egon Brunswik. His representational design and his lens model are full of principles of symmetry and we should make a try to capitalize on them.

--Fig. 7 about here—

Fig. 7 depicts hierarchical predictor and criterion models and one immediately sees, that predictability is at its maximum, when we relate both at the right level of aggregation and generality. Symmetry in amount of aggregation and symmetry in level of generality should boost predictability. Fishbein and Ajzen (1975) had been among the first to hint to these problems in attitude research and Ajzen (1988) in personality research. They coined these as principles of correspondence. They distinguished between single and multiple act criteria, pointed to the necessity of distinguishing between action, target, time and context and where able to demonstrate higher predictions when these elements corresponded to the criteria chosen.

The symmetry principles of the lens model also have been formulated in a mathematical equation derived by Tucker (1964), which can be used to explain successes and failures as regards prediction. We have augmented that equation with psychometric knowledge and after doing so it looks like this:

1) $r_{PR,CR} = S\,(r_{tt}^{PR}\, r_{tt}^{CR})^{0.5}\, G_{PR,CR}^{true}\, R_{PR}\, R_{CR} + e$

The correlation $r_{PR,CR}$ is the observed empirical correlation, $G_{PR,CR}^{true}$ is the true validity coefficient. The parameter S is mapping restriction or enhancement of range effects, and is:

2) $S = u\, G_{PR,CR}^{true} / (1+ (u\, G_{PR,CR}^{true})^2 - (G_{PR,CR}^{true})^2)^{0.5}$ and

3) $u = SD_{sample} / SD_{population}$

The term e is a random error component. The remaining two parameters R_{PR} and R_{CR} assess the overlap of the predictor and the criterion with the true construct in terms of multiple regression, they basically map the amount of symmetry between the predictor and the criterion model. In Fig. 6 this product $R_{PR}R_{cr}$ is one, for all cases of symmetry in level of generality. Once we use a narrow predictor and a broad criterion or vice versa, this product is considerable lower than one and consequently dramatically attenuates what we observe as the empirical relationship. Eq. 1 is typically used in meta-analysis and parameters S, e and the reliabilities are used to explain the variability in the observed distribution of correlation coefficients. The variability in the product, which is symmetry or correspondence was never used to my knowledge.

In personnel selection the ubiquity of general intelligence is widely discussed and accepted by many experts. The consensus in intellectual ability research is now hierarchical models. Amazingly the lower level abilities rarely demonstrate incremental validity. In terms of Brunswik symmetry we believe that this is an artifact of the criteria chosen, namely most often supervisor ratings. Supervisors probably pick up more or less random slices out of the incumbents performance and subjectively aggregate them, thus biasing their assessment towards a g-type criterion. So should we wonder, that g is the best predictor of such a criterion? In motivation theory we do not find such a consensus on a hierarchical structure. The construct of motivation is very fuzzy and consists of many domain specific facets. Can someone be motivated without clear goals, without concrete and specific interests, without any self-esteem or self-efficacy? Scanning the literature on research of motivation demonstrates this shifting and wandering emphasis. Correlating them with a g-type criterion suffers from asymmetry and results in low correlations.

<u>Are we looking at the wrong criteria?</u>

Another difference between can-do and will-do research is the trait-state distinction. In motivation and personality research we find the distinction between traits as dispositions, which are tendencies to act in a certain direction. A tendency can fairly be evaluated only after having a multiple act criterion assessed and aggregated over many situations and thus over a longer time-period. States are constructs, which are tailored to assess the situational specifity and to explain the variability of performance, thus traits should have higher symmetry to the level and states to variability of performance. Although the trait-state distinction could be considered in ability research as well, one rarely finds it. Can-do factors are mainly traits, whereas motivational constructs might be an unknown mixture of trait and state variance components.

Cattell's data box can be partitioned via split-plot designs into a between and a within source of variance. The former after factoring leads to trait factors and the latter one to state factors. Fig. 8 shows such a partitioning for the predictor and the criterion data box and we can ask how the traits and the states of the predictor box are related to the traits and the states of the criterion box. We see that there are four possibilities of relationships and that to test them we need assessments of the criterion over many situations and time points. We rarely have such an extensive assessment of the criterion.

--- Fig. 8 about here----

In personality research we applied years ago these distinctions in validating extraversion and emotional stability/lability (neuroticism) from the FPI against extensive self and peer reported data gathered over eight weeks. We constructed two multiple act criteria with indicators for extraversion derived from Eysenck's research and aggregated them over the eight weeks. This type of criterion can be considered as the mean level of performance. To construct an adequate criterion for emotional stability/lability it was evident that we could not use the same strategy, but instead we computed means for each half week and then the absolute differences and aggregated them over the fifteen scores derived this way. Whether behavioral indicators go up or down doesn't matter, but if it does it maps variability. Stability/lability should predict variability in mood and other indicators relevant for this construct. Fig. 9 demonstrates the impressive results in terms of convergent and discriminant validity. These results hint that we often might not be precise enough in our theoretical framework and use the wrong type of criteria in validating "will-do" constructs. This again complies with Kanfer and Ackerman's (2000) plea to make a distinction between proximal and distal effects of motivation as traits and as states.

- Fig.9 about here-

Ruth and Phil kindly allowed me to check some of our conceptual and methodological strategies at one of their TRACON data sets. It is known that the number of airplanes landed is the most cognitively demanding part of the task, which as level of performance can be well predicted from intellectual abilities. The TRACON criteria are also available over 30 time-points. The performance variability can be seen in Fig. 9a-b. Fig. 9a shows the individual trace lines over the 30 trials. In Fig. 9b five trials in one session are aggregated demonstrating the variability over the six sessions. A simple way to check whether it contains additionally to true variance in terms of stability, true sources of variability is to compute split-half and odd-even reliabilities as a function of amount of aggregation over the whole time-series. Fig. 10 shows this.

- Fig.9,9a-b and 10 about here—

In the lower curve we started estimating stability of performance via correlating point 1 with point 16, then we aggregated point 1 and 2 and correlated this we an aggregate of point 16 and 17. This procedure was done till the aggregate of all the first fifteen assessments had been correlated with an aggregate of the last fifteen performance data (number of airplanes landed). In the upper curve we correlated the first point with the second one and then build an odd aggregate by point 1 and 3 and correlated this with an even aggregate out point 2 and 4. This procedure was continued until an aggregate out of all odd points was correlated with an aggregate out of all even points. The odd-even curve estimates the reliability of performance in terms of true stability and true variability and the split-half one only in terms of true stability. We see that the two curve level off to a different plateau. The differences between the two plateaus map variability in performance as true individual differences in intra-individual change. The whole theory behind these concepts is given in Wittmann (1988) as a multivariate reliability and validity theory.

We assume that especially this source of variance might be better related to motivational constructs. As a simple approach to assess variability in performance we computed the standard deviation over number of airplanes landed over all 30 time-points. Unfortunately those who learn more and get to a higher plateau of performance also tend to have the larger SD. But we can hold constant the level of performance and check whether after this motivational constructs explain incremental variance on the partialed SD of that performance. Fig.11 shows that this is the case:

-Fig. 11 about here-

Admittedly the analysis was done exploratory and a little bit by snooping in the data. The interesting fact is, once we hold constant mean level of performance we can account for additional roughly 12% of the variability of performance. No gender differences any more and all the variables with increments are from the “will-do” camp. The higher mean self-efficacy the lower the variability. The higher variability in self-efficacy, the higher motivational skills and the higher interest in arts the higher the variability in performance.

All the intellectual abilities like math, spatial and perceptual speed, which very well predict level of performance, are not related to that variability. So depending on the type of criteria used, we get different messages for the impact of can-do and will-do factors.I also found another hint to the possible importance of will-do factors. Analyzing the TRACON data by split-plot test designs (repeated measurement analysis), I found the following gender paradox for the can-do factors. Fig. 12 shows the results, where the performance process for gender and g dichotomized at the overall mean is shown. We see large gender differences and g-type differences. The puzzling effect is that the smart girls with g above the mean are no better than the not so smart boys below mean g. The smart boys show the best and the not so smart girls the worst performance. Fig. 12 shows the intellectual ability profiles of all four groups, and the smart girls do not differ from the smart boys, nor do the not so smart girls differ from the not so smart boys, at least in terms of general ability g! The can-do factor g thus cannot explain these differences of the gender groups in performance. But we need more research what will-do factors could be used for explanation. My favorite candidate at the moment, bolstered by our own research, is risk-aversiveness. Girls risk less and thus inadvertently produce a learning environment with fewer opportunities to learn given the same level of aptitude as the boys who risk more.

- Fig. 12 about here-

Are we using the wrong measures?

Another problem is the dominance of self-report measures to assess the will-do factors. David McClelland and Heinz Heckhausen believed that motivation is better assessed via projective techniques like the TAT. Unfortunately projective techniques suffer from low reliability despite some improvement on standardization. In terms of the lens model equation the reliability product for them is consistently lower than the reliability product for self-report measures, but McClelland and Heckhausen believed that the construct validity product $R_{PR}R_{CR}$ is higher for the TAT assuming that it is closer to the construct. Theoretically at least, it is conceivable that the higher value of this product compensates for the lower reliability product. Thus:

$(\text{proj_}r_{tt}^{PR}r_{tt}^{CR})^{0.5}\ (\text{proj_}R_{PR}R_{CR}) > (\text{self_}r_{tt}^{PR}r_{tt}^{CR})^{0.5}\ (\text{self_}R_{PR}R_{CR})$, despite:

$\text{proj_}r_{tt}^{PR} < \text{self_}r_{tt}^{PR}$.

Under these circumstances the empirical validity $r_{PR,CR}$ observed for projective techniques indeed may be higher than those for self-report measures. McClelland actually reported a correlation over .30 with need for achievement and salary over a 20-30 year period.If one could improve on the TAT reliability by still better scoring rules and more extended assessments the pay off even may be larger. So give the TAT a second chance?

Psychometric problems in operationalizing the interaction term

The interaction hypothesis of ability and motivation is theoretically also very convincing. Interactions are psychology's favorite child, like in Vroom's theory or in the expectancy value models, but why don't they fulfill their promises? Again going back to the Brunswik lens model we get hints for explanations. This is especially true for studies where motivation was experimentally manipulated via reinforcements and other incentives. Substituting the predictor by the experimental treatment dummy, we ask what the reliability of the experimental treatment and what its construct validity is. In experiments, motivation as a causal variable is a dummy variable with a one for the experimental group and a zero for the control. Experimenters assume that this variable has perfect reliability; there is no variability within both groups. But this is only wishful thinking, often the reliability of these treatment variables is amazingly low, see Lösel and Wittmann (1989) for a demonstration. Additionally, dichotomizing a continuous dosage variable leads to loss of power (Cohen, 1983.). Multiplying a variable of low reliability, with one of high reliability (aptitudes as assessed in most tests surely have high ones), leads to a partialed product term assessing the interaction with low reliability as well.

Conclusions

Principles of symmetry and investing in more extended assessment of performance are essential in applying these conceptual and methodological proposals. Assessment of performance over longer time periods is rarely done, with the exception of the Kanfer/Ackerman lab. Had we only invested the same amount of energy in investigating and dimensionalizing work performance over the last 100 years, like we have done in the predictor area, I'm convinced, that the results would have unraveled a similar broad and differentiated hierarchical structure. With such knowledge available, we could fully test the promises of the principles of symmetry and how the different sources of true variance are related to each other. I'm convinced that the will-do factors would stand up and

demonstrate their importance. Industrial and organizational psychology should invest in solving that criterion problem and must exploit the new, powerful devices now being available with the so-called ambulatory assessment tools, which allow one to assess data in the real world, i.e. at the workplace over longer time periods, see Fahrenberg and Myrtek (1996) and Fahrenberg, Leonhart and Foerster (2002) for overviews and details. Despite other important aspects, the criterion problem is the most important one in organizational psychology and other areas as well and it is the neglected stepchild in most research. I agree with many points Frank Schmidt (2002) recently brought up in an issue in Human Performance and really admire his and Jack Hunter's work, but I completely disagree with him, that the criterion problem does not matter in explaining the ubiquity of g as the most prominent can-do factor. In a similar way I'm convinced that the will-do factors only will loose their disappointing status, once we better deal with the criterion and begin to fully understand aggregation and what effects are produced by it.

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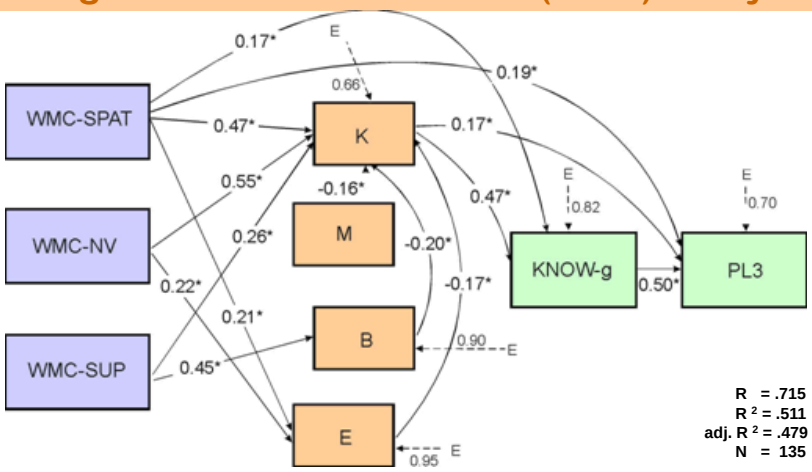
Fig.1 The Mannheim (1997) Study



Prediction and explanation of performance from the group-factor level
WMC-g = General factor of all working memory tasks;
BIS = Berlin intelligence structure;
KNOW-g = total knowledge;
PL3 = total computer games performance;
WMC-SPAT = Spatial working memory factor;
WMC-NV = Verbal-Numerical working memory factor;
WMC-SUP = Processing speed working memory factor;
K = reasoning; M = short-term memory; B = speed; E = creativity.

EQS Summary Statistics
Method: ML

Model CHI-Square	21.93
df = 22 p-value	0.4642
BBNFI	0.928
BBNNFI	1.000
CFI	1.000

Fig.2 The Berlin (1989) study

A nomological network for prediction and explanation of complex problem solving

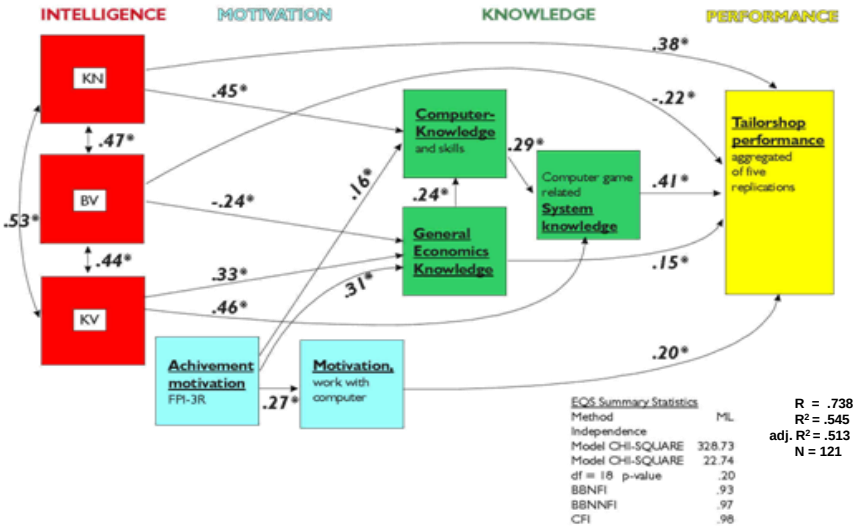
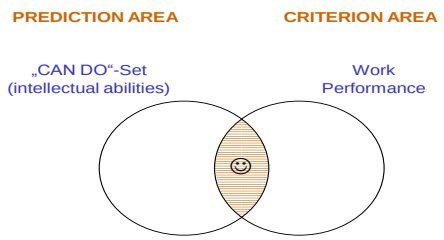
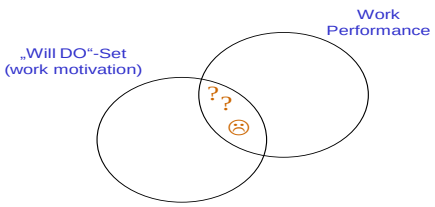


Fig.3



Meta-analysis demonstrates that intellectual abilities are the best predictors out of the „CAN-DO"-Set

Fig. 4



What about the „WILL DO"-Set ?

Fig. 5

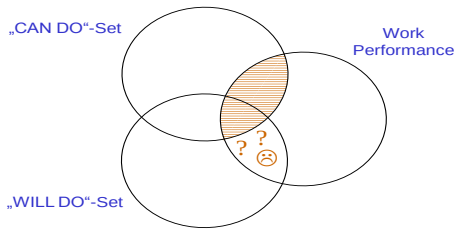


Fig. 6

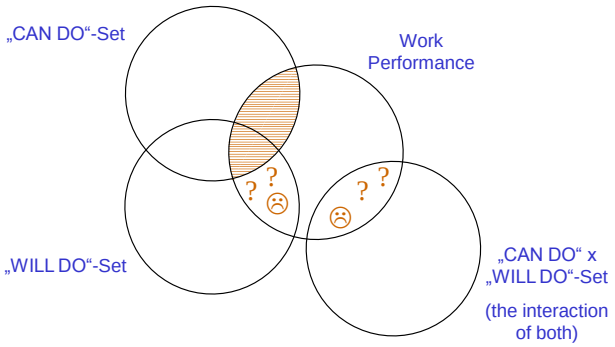


Fig.7 The true Brunswik-symmetrical latent structure of nature

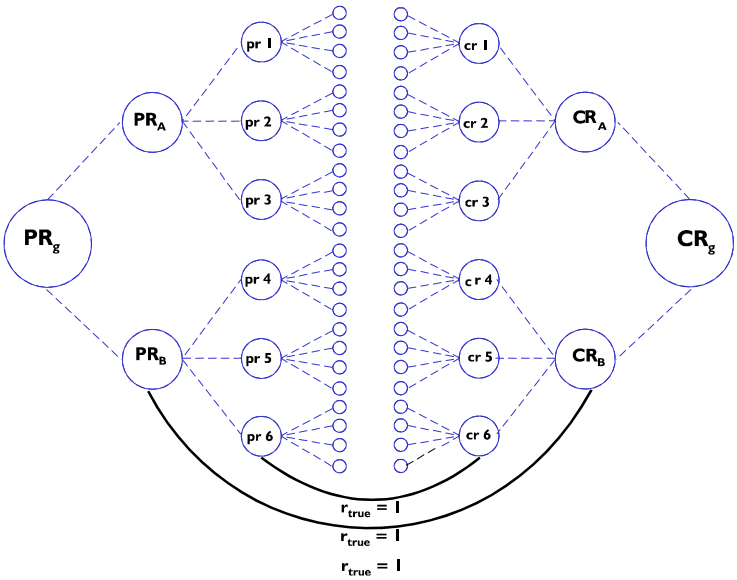
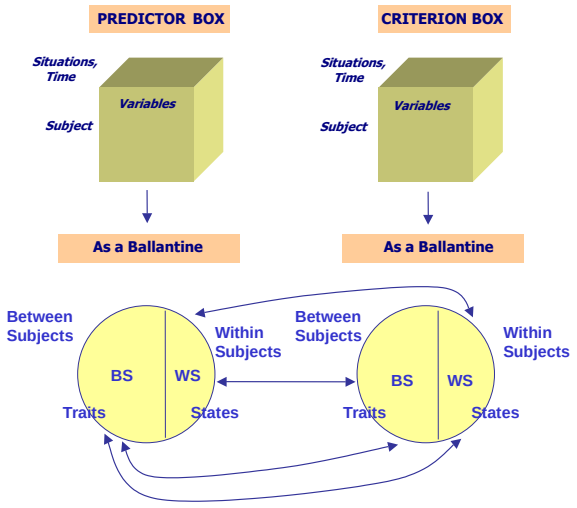


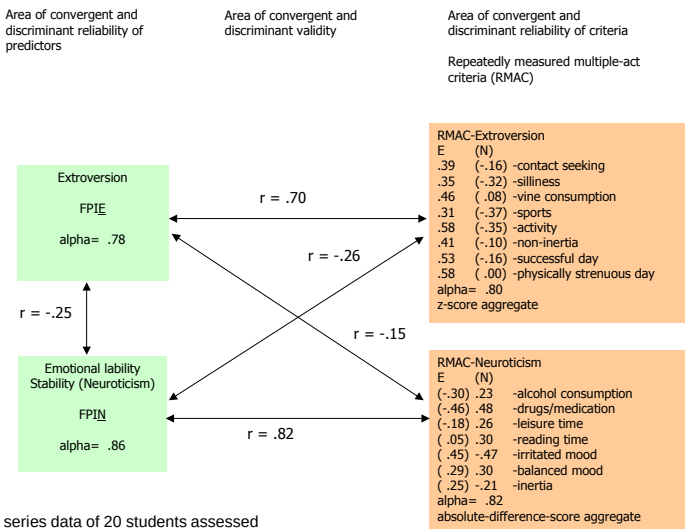
Fig. 8 Data-Box Partitioning
(Partitioning of Variance/Covariances)



$$F_G^{PR} = F_{BS}^{PR} + F_{WS}^{PR} \qquad F_G^{KR} = F_{BS}^{KR} + F_{WS}^{KR}$$

Mixture: Trait State Factors

Fig. 9 Testing Eysenck's E-/N-Theory in the Brunswik-symmetry framework [1]



[1] Time series data of 20 students assessed over 8 weeks from Fahrenberg et al. 1977

Fig.9a: Impressions about performance variability

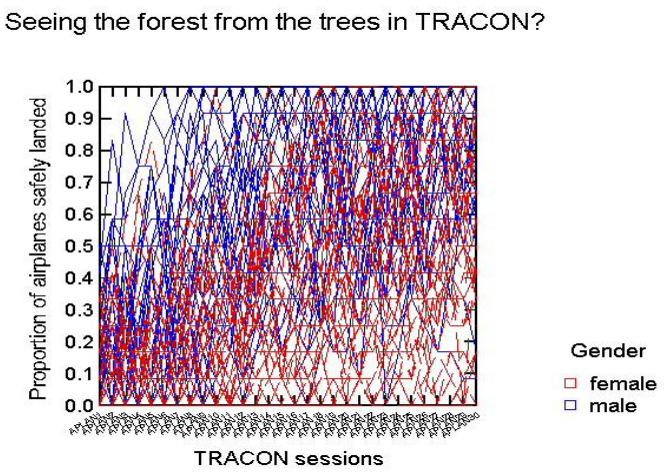


Fig. 9b Level and variability of performance

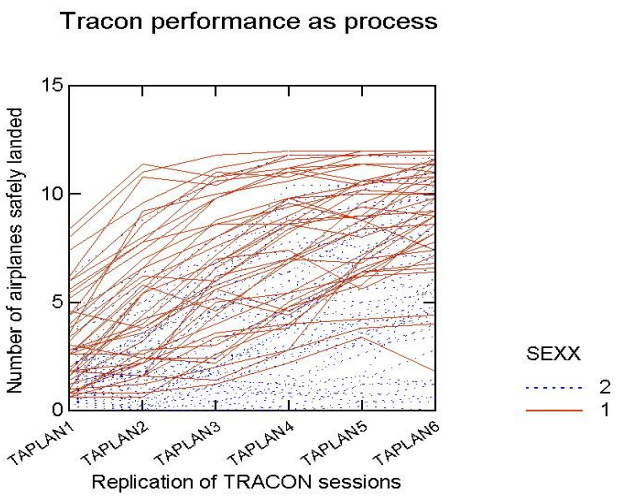


Fig. 10 Tracon performance: Planes arrived (Ackerman)

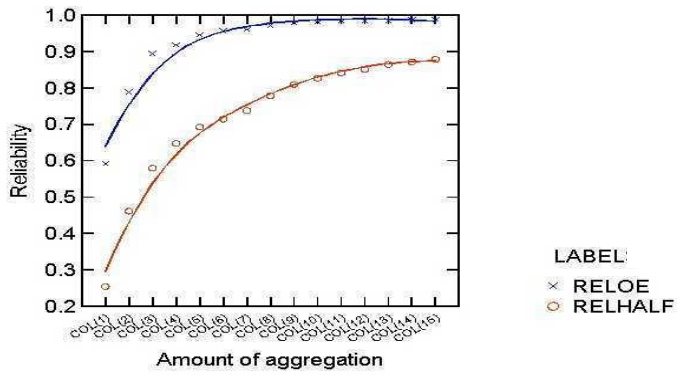


Fig. 11 Predicting performance variability in TRACON

Dep Var: SDAPLANE N: 93 Multiple R: 0.576 Squared multiple R: **0.332**

Adjusted squared multiple R: 0.325

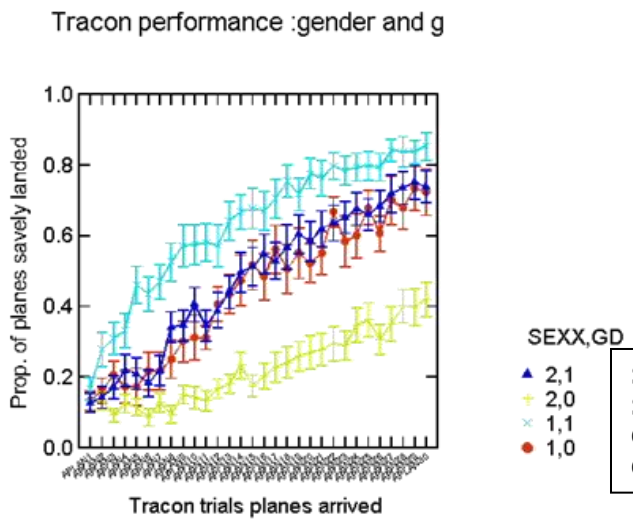
Effect	Coefficient	Std Error	Std Coef	Tolerance	t	P(2 Tail)
CONSTANT	0.133	0.012	0.0	.	10.871	0.000
TAPLANG	0.014	0.002	0.576	1.000	6.725	0.000

Dep Var: SDAPLANE N: 93 Multiple R: 0.674 Squared multiple R: **0.454**

Adjusted squared multiple R: 0.416

Effect	Coefficient	Std Error	Std Coef	Tolerance	t	P(2 Tail)
CONSTANT	0.034	0.045	0.0	.	0.746	0.458
TAPLANG	0.017	0.003	**0.685**	0.492	6.039	0.000
SELG	-0.003	0.001	**-0.382**	0.370	-2.913	0.005
SDSEL	0.007	0.002	**0.280**	0.812	3.168	0.002
MOTSKIL	0.001	0.000	**0.203**	0.806	2.293	0.024
ART	0.001	0.000	**0.173**	0.899	2.057	0.043
SEXX	-0.019	0.013	**-0.144**	0.647	-1.451	0.150

Fig.12 The gender puzzle



SEXX=1 male
SEXX=2 female
GD=0 intelligence below the mean
GD=1 intelligence above the mean

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